THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 8: **Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 32	substitute $n = 12$
2	**(d)** 81	substitute $n = 20$
3	**(c)** $n = 113$	$3n - 7 = 332 \Rightarrow n = 113$
4	**(a)** −21	$a + (n-1)d = 21 + 14 \times (-3) = -21$
5	**(a)** 67	$a + (n-1)d = 7 + 10 \times (6) = 67$
6	**(a)** 5050	$\frac{100 \cdot 101}{2}$
7	**(c)** 320	$ar^{n-1} = 5 \times (2)^6 = 320$
8	**(b)** 30	3, 6, 30
9	**(b)** a GP with $r = 2$	constant ratio 2
10	**(c)** n^2	$1 + 3 + 5 + \cdots + (2n-1) = n^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$t_1 = -1,\ t_2 = 2,\ t_3 = 5,\ t_4 = 8,\ t_5 = 11.$	1+1
12	$-3,\ -8,\ -13,\ -18,\ -23.$	1+1
13	$t_1 = 2,\ t_2 = 3,\ t_3 = 6,\ t_4 = 11,\ t_5 = 18.$	1+1
14	$t_{10} = 47$ and $t_{15} = 72.$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$3n - 7 = 332 \Rightarrow n = 113$ (1). $3n - 7 = 557 \Rightarrow 3n = 564 \Rightarrow n = 188$, a natural number, so 557 is the 188th term (2).	1+2
16	$5n - 2 = 308 \Rightarrow n = 62$: yes (1+). $5n - 2 = 471 \Rightarrow n = 94.6$, not a natural number ⇒ not a term (2).	1+2
17	$a + 2d = 12$, $a + 49d = 106$ ⇒ $47d = 94$, $d = 2$, $a = 8$ (2). $t_{29} = 8 + 28 \times 2 = 64$ (1).	2+1
	OR: 12, 15, …, 99: $99 = 12 + (n-1)3 \Rightarrow n = 30$ (1). Sum $= \frac{30}{2}(12 + 99) = 1665$ (2).	1+2
18	$500000 + 20000k = 600000 \Rightarrow k = 5$ raises (so he earns ₹6,00,000 in his 6th year).	2+1
	OR: $S_{25} = \frac{25 \times 26}{2} = 325$ marbles.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Write the sum forwards and backwards and add: $2S = n(n+1)$ (2). (ii) $S_{20} = 210$, $S_{50} = 1275$ (1). (iii) $S_{58} - S_{24} = 1711 - 300 = 1411$ (2).	2+1+2
	OR: (i) ₹240, ₹280, ₹320 (1). (ii) Constant difference 40; $t_n = 200 + 40n$ (2). (iii) ₹600 (1). (iv) $200 + 40n = 1000 \Rightarrow n = 20$ km (1).	1+2+1+1
20	(i) $S_{10} = 55$ (1). (ii) $S_{25} = 325$ (1). (iii) $\frac{n(n+1)}{2} = 210 \Rightarrow n = 20$ (2). (iv) The 10th triangular number (1).	1+1+2+1
	OR: (i) 1, 3, 9, 27 (1). (ii) 81, 243 (1). (iii) 3^n, a GP with $r = 3$ (1). (iv) $\frac{27}{64}$; $\left(\frac{3}{4}\right)^n$ (1). (v) It decreases towards 0, while the number of triangles increases rapidly (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapter 8: **Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** −28	substitute $n = 6$
2	**(a)** −23	substitute $n = 15$
3	**(c)** $n = 25$	$4n - 3 = 97 \Rightarrow n = 25$
4	**(a)** −13	$a + (n-1)d = 11 + 8 \times (-3) = -13$
5	**(c)** 155	$\frac{10}{2}(2 \cdot 2 + 9 \cdot 3) = 155$
6	**(d)** 180	$\frac{8}{2}(10 + 35) = 180$
7	**(b)** 108	$ar^{n-1} = 4 \times (3)^3 = 108$
8	**(b)** 21	13+8=21
9	**(a)** neither an AP nor a GP	differences 3,5,7; ratios 4, 2.25, …
10	**(c)** common ratio	definition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5n - 3 = 97 \Rightarrow n = 20$ — yes (20th term). $5n - 3 = 172 \Rightarrow n = 35$ — yes (35th term).	1+1
12	$5n - 3 = 607 \Rightarrow 5n = 610 \Rightarrow n = 122$; the 122nd term.	1+1
13	$-5, -2, 1, 4, 7$ (1). $t_n = -5 + 3(n-1) = 52 \Rightarrow n = 20$, so 52 is the 20th term (1).	1+1
14	$T_4 = 7,\ T_5 = 13,\ T_6 = 24.$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$S_{40} - S_{11} = \frac{40\times41}{2} - \frac{11\times12}{2} = 820 - 66 = 754.$	2+1
16	Last term $= 3 + 19 \times 5 = 98$ (1). Sum $= \frac{3+98}{2} \times 20 = 1010$ (2).	1+2
17	$d = 3$: $t_n = 1 + (n-1)3 = 3n - 2$ (2). Recursive: $t_1 = 1,\ t_n = t_{n-1} + 3$ (1).	2+1
	OR: $t_n = \frac{n(n+1)}{2}$: $t_{10} = 55$, $t_{17} = 153$, $t_{80} = 3240$.	1+1+1
18	$1 + 3 + 5 + 7 + 9 = 25$ ✓ (1). For the AP $1, 3, \ldots, (2n-1)$: $S = \frac{n}{2}(1 + (2n-1)) = n^2$ (2).	1+2
	OR: $t_{10} = 3 \cdot 2^9 = 1536$ (2). $t_1 = 3,\ t_n = 2t_{n-1}$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $18,\ 13.5,\ 10.125,\ 7.59375,\ 5.695$ (2). (ii) $t_n = 24\left(\frac{3}{4}\right)^n$ (1). (iii) After 7 bounces 3.20 ft; after the 8th 2.40 ft ⇒ 8 bounces (2).	2+1+2
	OR: (i) 5,00,000; 5,20,000; 5,40,000 (1). (ii) $500000 + 7 \times 20000 = ₹6,40,000$ (1). (iii) $500000 + 20000(n-1) = 800000 \Rightarrow n = 16$ (1). (iv) $\frac{10}{2}(2 \cdot 500000 + 9 \cdot 20000) = 5 \times 1180000 = ₹59,00,000$ (2).	1+1+1+2
20	(i) $a + 2d = 12,\ a + 49d = 106 \Rightarrow d = 2,\ a = 8$ (2). (ii) $8 + 28 \times 2 = 64$ (1). (iii) $\frac{50}{2}(8 + 106) = 2850$ (2).	2+1+2
	OR: (i) $1, 2, 3, 5, 8, 13, 21, 34, 55, 89$ (2). (ii) $1 + 2 + 3 + 5 + 8 + 13 + 21 + 34 = 87 = V_{10} - 2$ (2). (iii) By Virahānka, 7th century CE, in connection with Prakrit metre (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.